

Quantum Systems in Cavities

Spectral properties of an SSH chain coupled to a cavity mode

Valentina C. Barroso, Pablo S. Cornaglia, Gonzalo Usaj



Floquet Engineering vs. c-QED

Floquet Engineering:

- Uses external time-periodic driving (e.g., lasers) to tailor topological properties.
- *Drawback:* Inherently out-of-equilibrium (heating, uncontrolled level population).

Cavity QED (c-QED):

- Strong light-matter coupling using vacuum fluctuations.
- *Advantage:* Access to the equilibrium properties of the 'polaritonic' system.

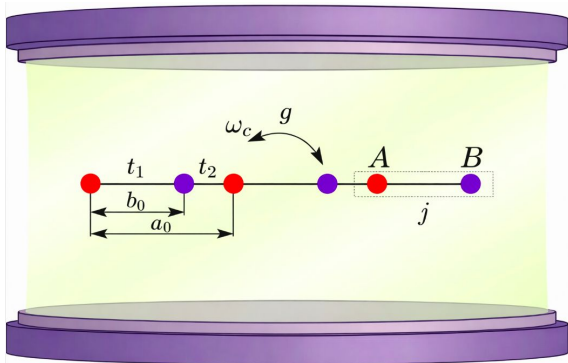


The Model: SSH Chain in a Cavity

Hybrid System Hamiltonian:
$$H = \tilde{t}_1 \sum_{j=1}^L \left(c_{jA}^\dagger c_{jB} + \text{h.c.} \right) + \tilde{t}_2 \sum_{j=1}^{L-1} \left(c_{j+1,A}^\dagger c_{jB} + \text{h.c.} \right) + \omega_c \left(a^\dagger a + \frac{1}{2} \right)$$

The hopping amplitudes $\tilde{t}_1$ and $\tilde{t}_2$ are modulated by the boson field.

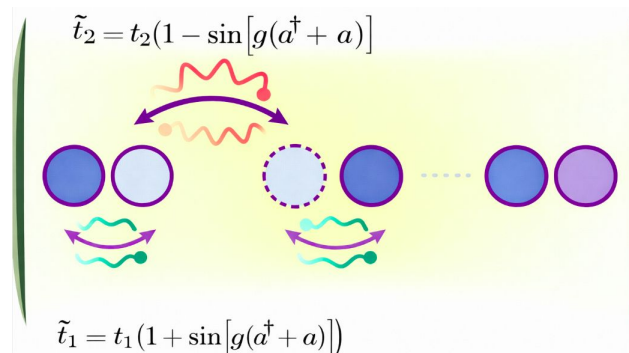
Peierls Substitution Coupling:



$$\tilde{t}_1 = t_1 e^{i \frac{g}{\sqrt{L}} b_0 (a + a^\dagger)}$$

$$\tilde{t}_2 = t_2 e^{-i \frac{g}{\sqrt{L}} (1-b_0)(a + a^\dagger)}$$

Phonon Modulation:



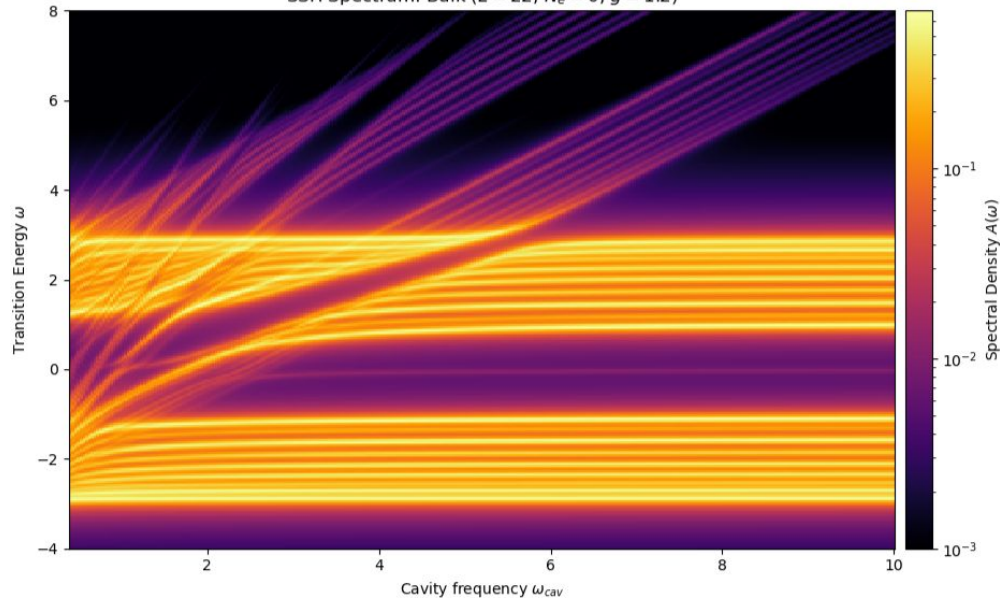
$$\tilde{t}_1 = t_1 (1 + \sin [g(a^\dagger + a)])$$

$$\tilde{t}_2 = t_2 (1 - \sin [g(a^\dagger + a)])$$

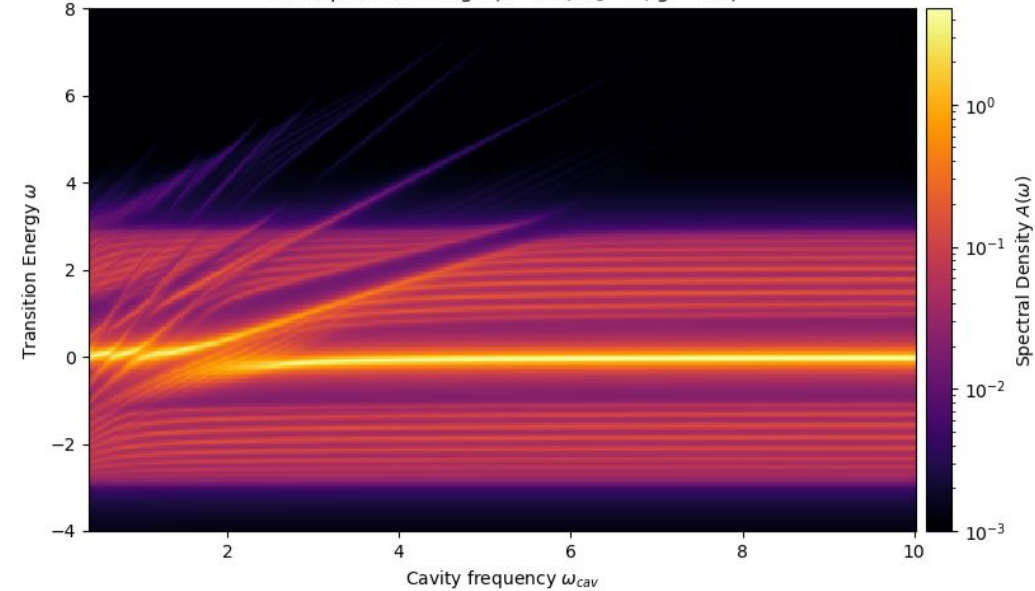


Peierls Substitution Coupling

SSH Spectrum: Bulk ($L = 22$, $N_e = 0$, $g = 1.2$)



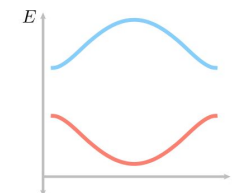
SSH Spectrum: Edge ($L = 22$, $N_e = 0$, $g = 1.2$)



Results for the Finite Chain (Empty Fermi Sea):

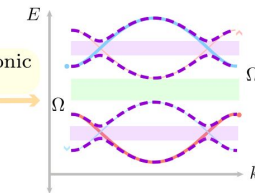
- A π gap opens up in the spectrum due to light-matter coupling.
- **No anomalous edge states** are observed within the induced gaps.

Electronic band structure



+ photonic field

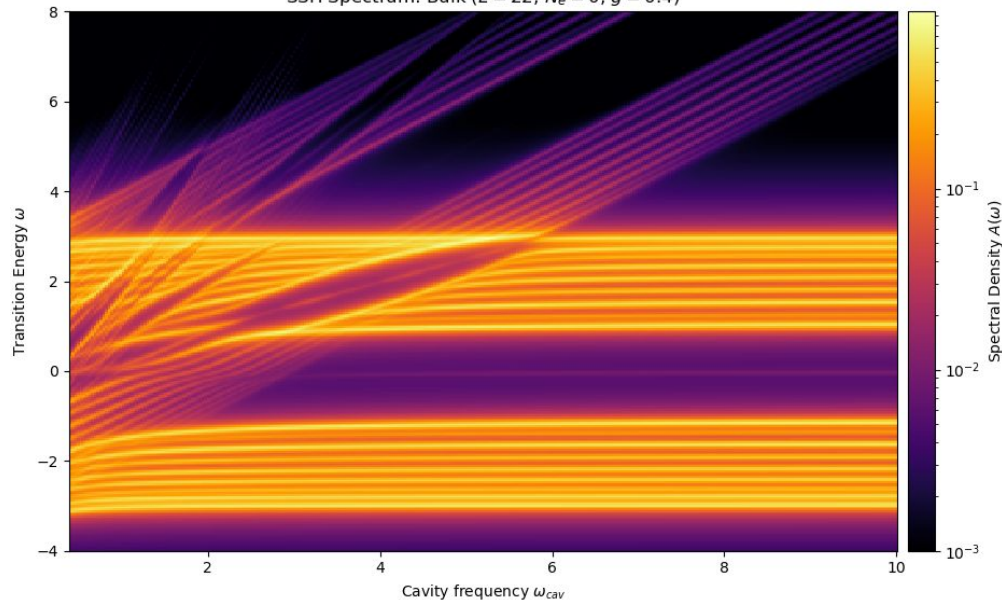
Hybridized band structure



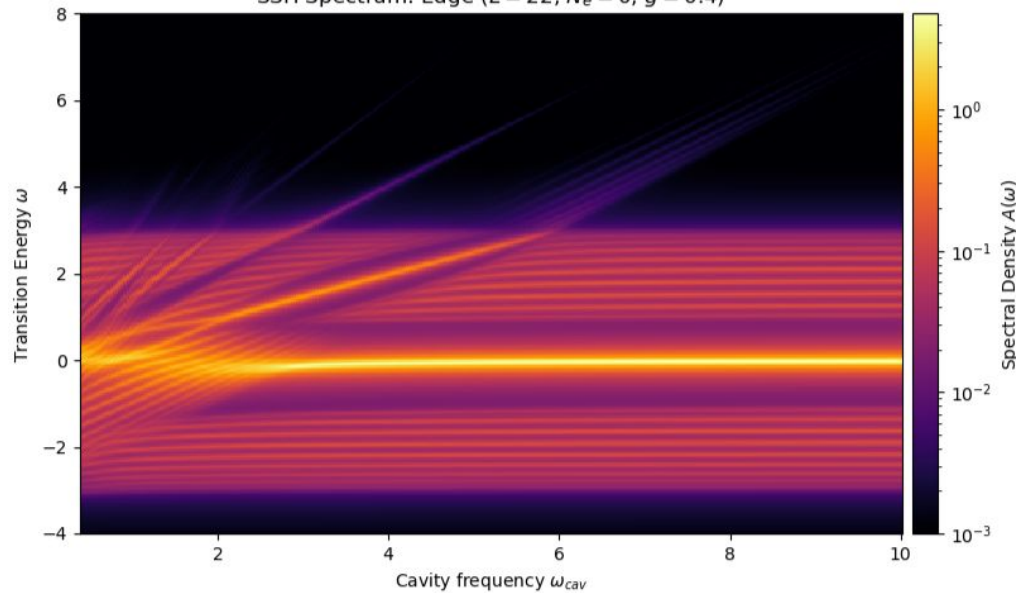


Phonon-like Coupling

SSH Spectrum: Bulk ($L = 22$, $N_e = 0$, $g = 0.4$)



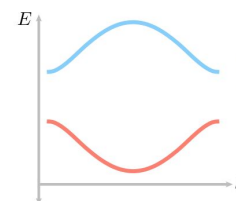
SSH Spectrum: Edge ($L = 22$, $N_e = 0$, $g = 0.4$)



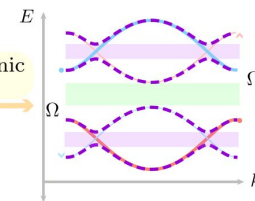
Results for the Finite Chain (Empty Fermi Sea):

- The photonic field hybridizes the electronic band structure.
- The **anomalous edge state** is clearly visible.

Electronic band structure



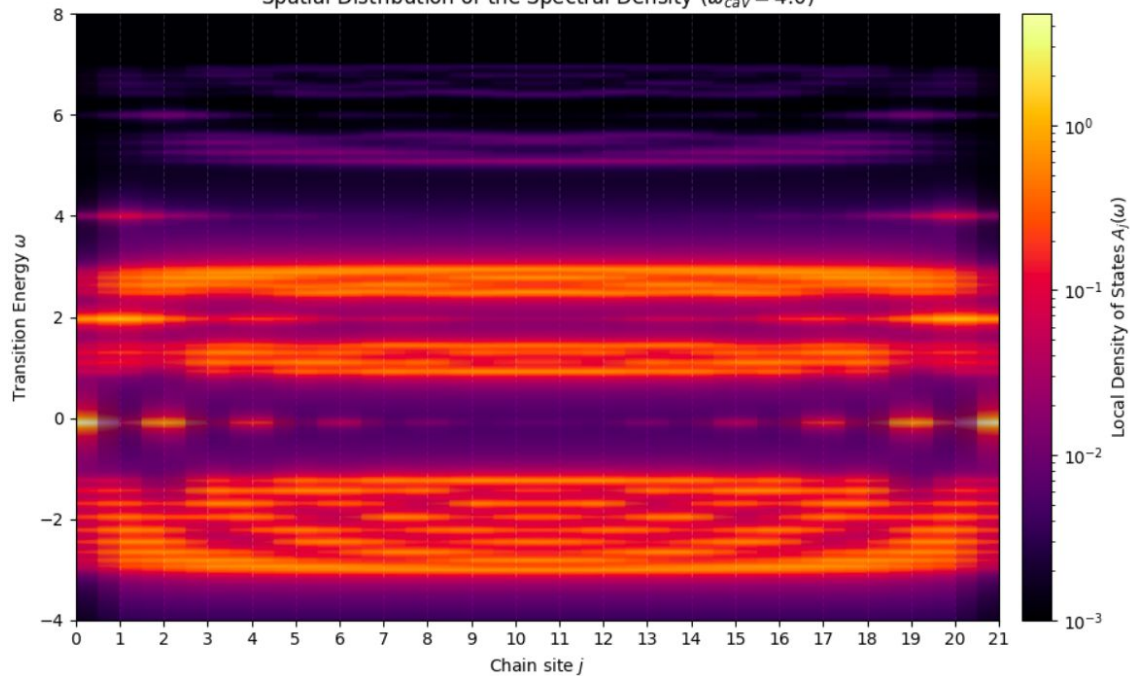
Hybridized band structure



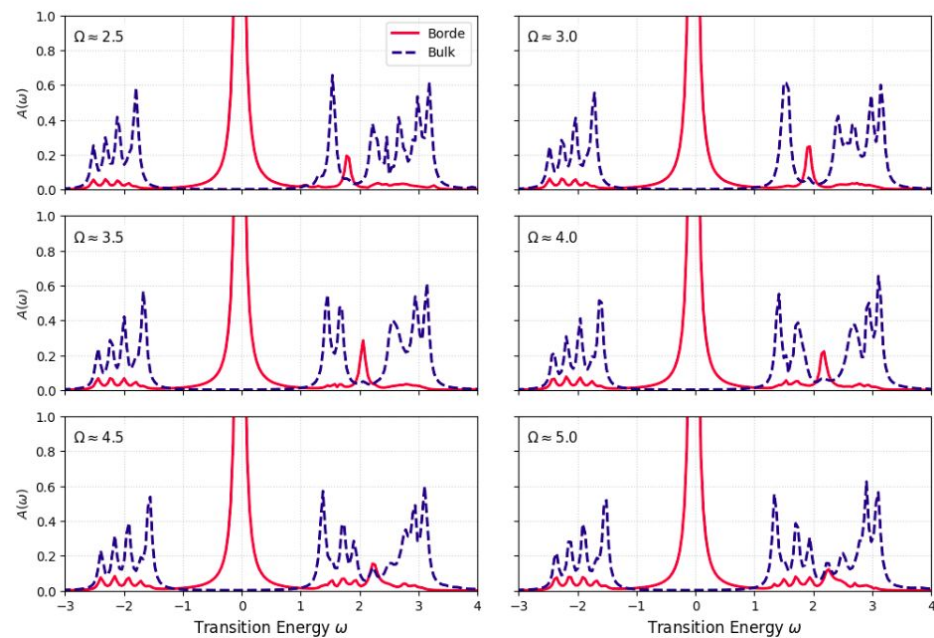
+ photonic field

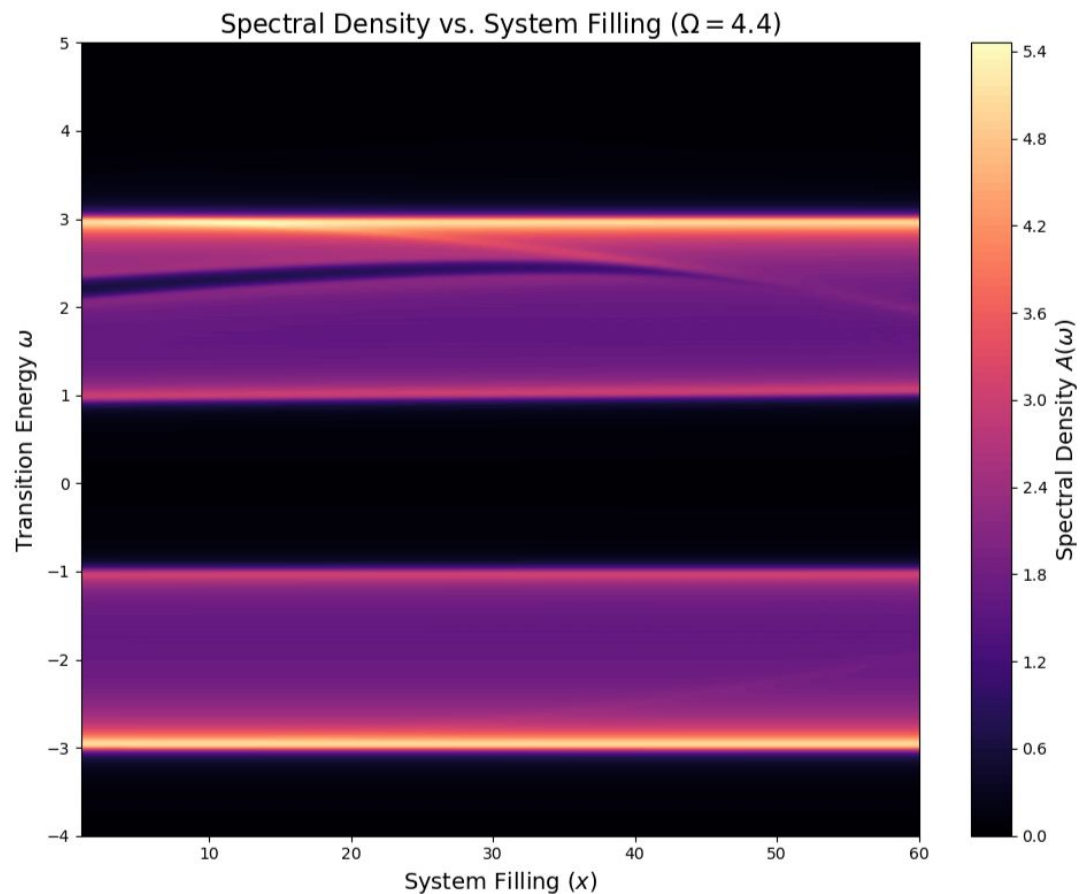


Spatial Distribution of the Spectral Density ($\omega_{cav} = 4.0$)



Spectral Density Comparison: Edge vs. Bulk ($L=22, N=5$)

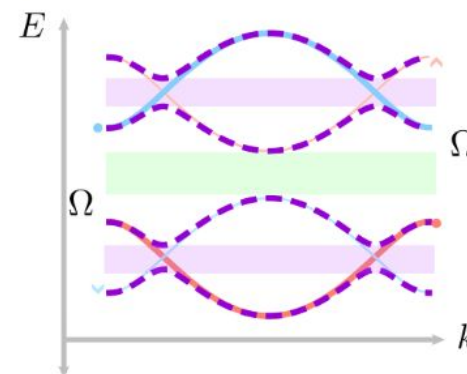




Critical Condition: Gap collapse occurs at

$$\omega \approx 2E_F$$

Hybridized band structure





Conclusions

- **Phonon-like vs. Peierls:** Anomalous edge states emerge under phonon-like coupling but are absent in the Peierls substitution model.
- **Quantum Floquet Analogy:** Isolated cavity-QED systems successfully mimic the physics of periodically driven (Floquet) materials.
- **Many-Body Constraints:** The induced π -gap is sensitive to electron density, closing as the cavity frequency ω approaches the $2E_F$ threshold.
- **Hybrid Nature:** Observed edge modes are strictly **light-matter hybrids**, differing spatially from standard SSH states.